TASK 10 – COMPARE EXPECTATIONS WITH EXECUTION

1. Aim

To compare floating-point calculations with Decimal calculations, measure execution time for different numbers of payments, examine rounding differences and evaluate the use of a dictionary as a rate table.

2. Introduction

The milk co-operative calculates payments for many farmers. The use of floating-point numbers can produce small precision errors. The Decimal data type is suitable for money calculations because it represents decimal values more accurately.

This experiment compares both methods using 300, 6,000 and 150,000 payments.

3. Float and Decimal Comparison

The same payment records are added using float and Decimal.

Number of payments	Expected Decimal total
300	₹77,400.00
6,000	₹15,48,000.00
150,000	₹3,87,00,000.00

The float total may contain a small precision difference because floating-point numbers use binary representation. Decimal maintains exact decimal values when created from strings.

The actual difference is calculated by subtracting the Decimal total from the float total. The Python program prints the measured difference for each size.

4. Execution Time Comparison

The program uses time.perf_counter() to measure how long each calculation takes.

The experiment is performed for:

- 300 payments
- 6,000 payments
- 150,000 payments

The actual execution time depends on the computer and Python environment. The measured times are recorded from the program output.

5. Rounding Each Payment vs Rounding Only the Total

There are two approaches:

Method 1: Round each payment

Each farmer's payment is rounded to two decimal places before adding it to the total.

Method 2: Round only the total

All unrounded payment values are added first. The final total is rounded to two decimal places.

For example, consider three payments of ₹10.005 each:

- Rounding each payment: ₹10.01 × 3 = ₹30.03.
- Rounding only the total: ₹30.015 becomes ₹30.02 using half-up rounding.

The difference is ₹0.01.

For a co-operative that pays each farmer separately, rounding each fa